STAT/MA 41600

In-Class Problem Set #37: November 13, 2026

1. As in Problem Set 18, Question 3, suppose that the number of raindrops that land on a detector during a certain fixed period of time has a Poisson distribution. If the mean number of raindrops during that period is 300, find an approximation for the probability that there are 295 or more raindrops during that period.

2. As in Problem Set 33, Question 5, upon drinking a witches' brew, each person will begin to feel the effects of the potion in an exponential time with mean 2 minutes (regardless of the type of witches' brew). Suppose that 400 people each simultaneously drink one of the witches' brews, and their times to feel the effects of the potions are independent random variables X_1, X_2, ..., X_{400}.

Define $X = X_1 + \cdots + X_{400}$.

2a. What type of random variable is X?

2b. What are the parameter(s) of X?

2c. Is the distribution of X approximately Normal? If so, what are the parameters of such a Normal distribution? If not, then why not?

2d. Find a good approximation for $P(X > 815)$.

3. A toy company sells boxes of bears. In each box, there are 7 bears, namely, 1 red, 1 orange, 1 yellow, 1 green, 1 blue, 1 purple, and then the 7th (bonus) bear in the box is an extra bear with one of the 6 earlier colors, with all 6 such colors equally likely for that bonus bear. The colors of the (7th) bonus bears are independent from box to box.

In a certain elementary school, all 500 students receive a box of 7 bears during a holiday celebration. Find a good approximation for the probability that there are *strictly more than* 580 blue bears altogether.

4. As in Problem Set 17, Question 3, consider a 6-sided dice with 1 red side, 2 blue sides, and 3 yellow sides. Catherine rolls the dice repeatedly, until the red side appears for the 50th time, and then she stops. Let X denote the number of blue results during this process. Find a good approximation for $P(97 \leq X \leq 103)$.

5. Consider a Poisson random variable Y with $E(Y) = 100$. Also consider the random variable X from Question 4, and assume that X and Y are independent. Find a good approximation for $P(|X - Y| \leq 10)$.